Morning Paper Report - Tuesday, September 08, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Zooming in on what counts as core and auxiliary: A case study on recognition models of visual working memory

Maria M. Robinson, J. Williams, J. Wixted, Timothy F. Brady
Psychonomic Bulletin & Review, 2024

Source: Semantic Scholar; matched terms: visual working memory, working memory, computational model, modeling

Link: https://www.semanticscholar.org/paper/2fb6f4b76b97f0516ed58b61c5b6fe71ae4da0e8

Goal: Research on best practices in theory assessment highlights that testing theories is challenging because they inherit a new set of assumptions as soon as they are linked to a specific methodology

Method: We focus on these issues in a reanalysis of a seminal study and its replications, both of which use a simple working-memory paradigm and a mainstream computational modeling approach

Results: We show that tracking auxiliary assumptions is difficult because they are made at different stages of theory testing and at multiple levels of a theory

Conclusion: Together, our work demonstrates why tracking and testing auxiliary assumptions remains a fundamental challenge, even in prominent studies led by careful, computationally minded researchers Our work also serves as a conceptual guide on how to identify and test the gamut of auxiliary assumptions in theory assessment, and we discuss these ideas in the context of contemporary approaches to scientific discovery.

2. The Missing Piece of the Puzzle: Predictive templates guide visual search by prompting the generation of target features in visual working memory.

J. Bartlett, Blaire Dube
Journal of Vision, 2025

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: https://www.semanticscholar.org/paper/b93f8928a7f89dc4de3be3e6ed4825b5f90b501e

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

3. Inhibition continues to guide search under concurrent visual working memory load

Zachary Hamblin-Frohman, Stefanie I. Becker
Journal of Vision, 2022

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: https://www.semanticscholar.org/paper/6da43a117f904eb151c9e564b08cfdf6be393a69

Goal: It is well known that attention can be automatically attracted to salient items

Method: In Experiment 1 the presence of a distractor facilitated visual search under low and high VWM loads, as reflected in faster response times when the distractor was present (compared to absent), and in fewer eye movements to the salient distractor than the non-target items

Results: However, recent studies show that it is possible to avoid distraction by a salient item (with a known feature), leading to facilitated search

Conclusion: This suggests that although VWM plays a role in guiding attention to the (irrelevant) target color, distractor-feature inhibition can operate independently.

4. Attentional enhancement predicts individual differences in visual working memory under go/no-go search conditions

Daniel Tay, J. McDonald
PLoS Biology, 2022

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: https://www.semanticscholar.org/paper/ac72ba26b977cbe774c870b7a44f1291ba091137

Goal: Attention-control processes transfer relevant information to visual working memory (WM) and prevent irrelevant information from consuming WM resources

Method: Here, participants searched for a pop-out target on Go trials and withheld responses on an equal number of randomly intermixed No-Go trials, depending on the colour of the stimulus array

Results: Although event-related potentials (ERPs) have revealed attention-control processes associated with enhancement of relevant stimuli (targets) and suppression of irrelevant stimuli (distractors), only the suppressive processes have been found to predict WM capacity

Conclusion: These findings indicate that target-enhancement mechanisms control access to WM in search tasks that require dynamic control and disconfirm the view that the gateway to WM is entirely inhibitory by nature.

5. Working memory relates to individual differences in speech category learning: Insights from computational modeling and pupillometry

Jacie R. McHaney, Rachel Tessmer, Casey L. Roark, B. Chandrasekaran
bioRxiv, 2021

Source: Semantic Scholar; matched terms: working memory, computational model, modeling

Link: https://www.semanticscholar.org/paper/6fdd0e28008078ceca85a70f714da91245044236

Goal: Across two experiments, we examine the relationship between individual differences in working memory (WM) and the acquisition of non-native speech categories in adulthood

Method: Across two experiments, we examine the relationship between individual differences in working memory (WM) and the acquisition of non-native speech categories in adulthood

Results: In Experiment 1, we show that individuals with higher WM acquire non-native speech categories faster and to a greater extent than those with lower WM

Conclusion: We propose that higher WM may allow for greater stimulus-related attention, resulting in more robust representations and optimal learning strategies We discuss implications for neurobiological models of speech category learning.